



Code-Signing Tool – HSM

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Rev. 3.0.1

May 2018



Revision Sheet

Release No.	Date	Revision Description
Rev. 0	24/05/2018	Initial Work

USER'S GUIDE

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1.0 GENERAL INFORMATION

This document provides the information necessary for the user to effectively use Code-Signing Tool with Hardware Security Module backend. It is primarily intended for users who are familiar with CST tool to sign codes for the NXP High Assurance Boot (HAB).

A. INSTALLATION

1.1 Dependencies

The CST-HSM backend depends on:

- OpenSSL 1.0.2x (required by the Frontend)
- Libconfig

Make is required for building the software.

- 1) Install OpenSSL

libconfig is C/C++ library for processing structured configuration files. It is used by CST to load HSM related configuration.

- 2) Install libconfig

1.2 Prebuilt binaries

Pre-compiled CST and HSM backend can be found on the delivered package.

Unzip the CST-HSM package.

For Linux 32-bits cst binary can be found under cst-hsm-3.0.1/release/linux32/bin

For Linux 64-bits cst binary can be found under cst-hsm-3.0.1/release/linux64/bin

1.3 Compile source

To compile backend and CST from source:

1.3.1 Compile backend

Compile the backend source code using the following command:

Options:

OPENSSL_PATH Specify prefix of path of OpenSSL

1.3.1 Compile Code-Signing tool

Compile CST using the following command:

**2.0 DEPLOY
Code-Signing
Tool**

B. DEPOLY

This section provides instructions and prerequisites to deploy Code-Signing Tool with 2 HSMs, these are SoftHSM2 and Utimaco HSM simulator.

2.1 Install HSM

2.1.1 SoftHSM2

Install SoftHSM2

Use either softhsm2-util to initialize a token. The SO PIN can e.g. be used to re-initialize the token and the user PIN is handed out to the application so it can interact with the token.

Type in SO PIN and user PIN.

Initialized tokens will be reassigned to another slot (based on the token serial number). It is recommended to find and interact with the token by searching for the token label or serial number in the slot list / token info.

2.1.2 Utimaco HSM Simulator

Download and extract SecurityServerEvaluation package.

Copy cs_pkcs11_R2.cfg to /etc

Edit **/etc/cs_pkcs11_R2.cfg** configuration file and

- Uncomment Logpath = /tmp in [Global] section
- Add Device = [3001@127.0.0.1](#) to [CryptoServer] section
- Save the file

Install java if its is not installed on your system

Launch p11cat

From the menu bar click **Login/Logout**

Select a slot from the **Slot List grid**

Select **Login Generic**

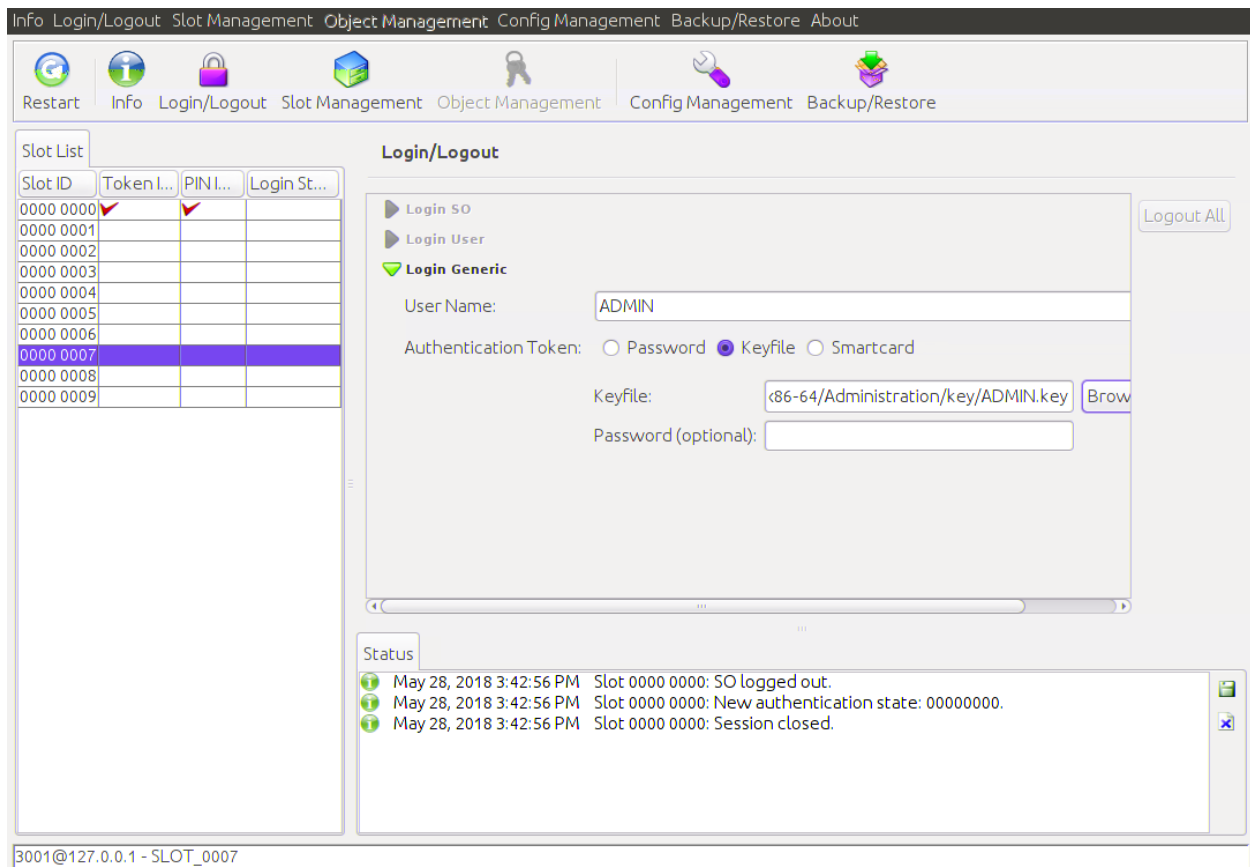
Type **ADMIN** as UserName

Select Keyfile as Authentication Token

Click browse button and fin the **ADMIN.key** file

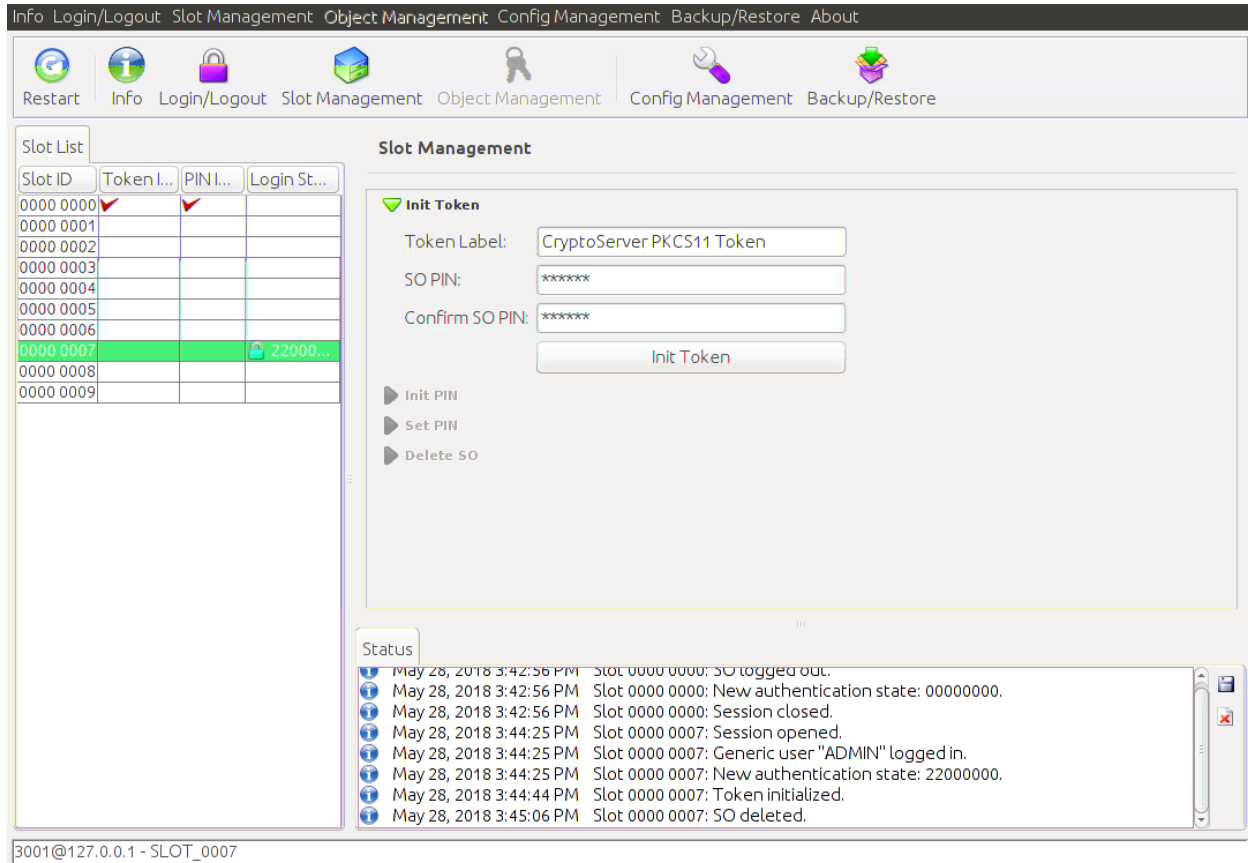
The keyfile is under SecurityServerEvaluation-V4.20.0.4/Software/Linux/x86-64/Administration/key/ADMIN.key

Click Login



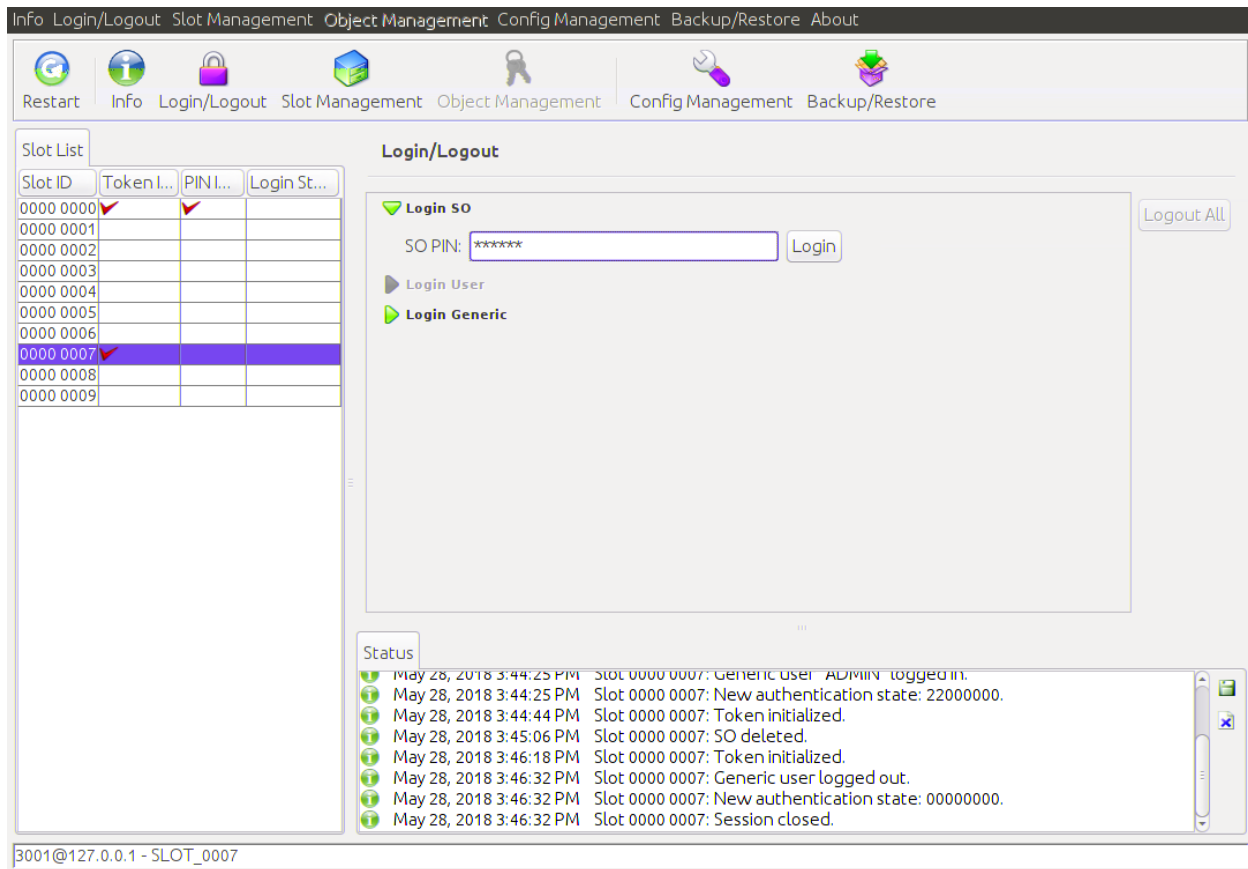
From **Slot Management** menu click

In **Init Token** form define the **Token Label** and **SO PIN** then click **Init Token**

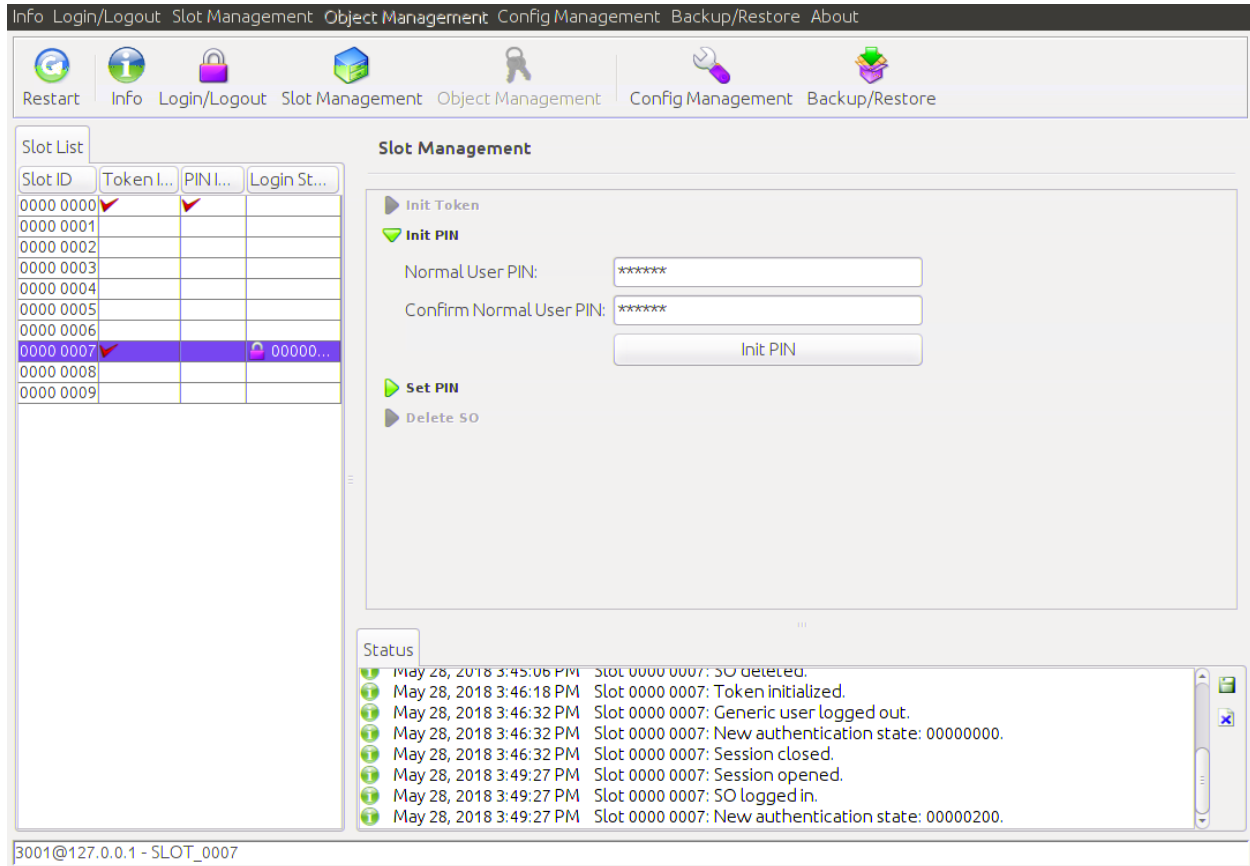


From **Login/Logout** menu click **Logout All**

From **Login/Logout** menu click **Login SO**



From **Slot Management** menu click **Init PIN**
Set User PIN for example 123456



2.3 Create certificates & keys

Referring to **Key and Certificate Generation HAB Code Signing Tool User's guide**, generate keys and certificates which will be used to sign boot loader image.

If you already have the cryptographic material on the disk please go directly to *push cryptographic material to HSM* section

2.3.1 Create serial file

serial - 8-digit OpenSSL uses the contents of this file for the certificate serial numbers.

2.3.2 Create passphrase file

key_pass.txt - Contains your pass phrase that will protect the HAB code signing private keys.

2.3.3 Generate the PKI tree

When prompts please answer according to the following:

Private keys will generate on keys directory and corresponding Certificates are placed in the crts directory.

2.3.4 Generate SRK table

SRK_1_2_3_4_table.bin - SRK table contents with HAB data
SRK_1_2_3_4_fuse.bin - contains SHA256 result to be burned to fuse

2.3.4 Command Sequence File (CSF)

CST tool requires CSF script. This file describes certificates, keys and the data ranges used in sign and encryption functions. Create **u-boot.csf** file.

2.4 Push cryptographic material to HSM

2.4.1 Install p11tool

To perform interact with HSM you can use p11tool.

2.4.2 Find Token URL

You need to locate your HSM vendor's PKCS#11 interface implementation in order to use it with p11tool and CST later.

Utimaco pkcs#11 library is *SecurityServerEvaluation-V4.20.0.4/Software/Linux/x86-64/Crypto_APIs/PKCS11_R2/lib/libcs_pkcs11_R2.so*

SoftHSM2 pkcs11# library is */usr/lib/softhsm/libsofthsm2.so*

Find the Token URL to interact with the token.

Example using SoftHSM2

In this case Token URL is

pkcs11:model=SoftHSM%20v2;manufacturer=SoftHSM%20project;serial=9c1aeb00e05a348b;token=CST-HSM

2.4.3 Push CSF certificate & key

To match a certificate to a private key, both should have the same ID, for some other HSM implementations they should have the same label.

Utimaco HSM Simulator

Private Key

Certificate

SoftHSM2

Private Key

Certificate

2.4.4 Push IMG certificate & key

Utimaco HSM Simulator

Private key

Certificate

SoftHSM2

2.4.5 Verification

Verify that all cryptographic materials were written correctly to token by running.

Enter PIN when prompts

2.5 Pull certificates from HSM

2.5.1 Install p11tool

To perform interact with HSM you can use p11tool.

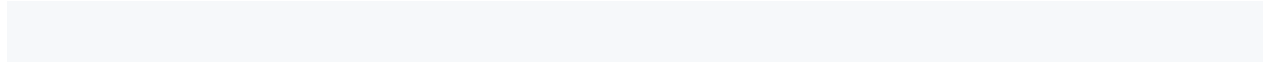
2.5.1 Find Token URL

Find the Token URL to interact with the token.

In this case Token URL is
`pkcs11:model=SoftHSM%20v2;manufacturer=SoftHSM%20project;serial=9c1aeb00e05a348b;token=CS
T-HSM`

2.4.3 Pull CSF & IMG certificate

List all certificates on HSM



Pull CSF certificate from HSM

Pull IMG certificate from HSM



